Probability from Observation:
Carathéodory and Stone Routes to the Observable Extension Theorem

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Abstract

We prove that a coherent family of observations determines a unique probability measure on the observable σ-algebra. The setting is a directed system of measurable outcome spaces connected by surjective refinement maps, together with a compatible family of finitely additive charges on the resulting cylinder algebra. We identify the necessary and sufficient condition for the family to extend to a σ-additive global measure: *collective exhaustion* (CE), the condition that no level can assign persistent mass to events the system as a whole sees as empty.

The argument proceeds by two independent routes, each illuminating a different aspect of the extension problem. The *Carathéodory route* assumes σ-additive marginals and constructs the global measure by a direct Carathéodory argument on the realization space, with no topological hypotheses on the outcome spaces. The *Stone route* assumes only finite additivity and derives σ-additivity from the compactness of the Stone space: the directed system of Boolean algebras dualises to an inverse system of compact Hausdorff spaces, whose inverse limit carries a canonical σ-additive measure; CE then ensures this measure concentrates on the image of the sample space inside its Stone compactification. Under shared hypotheses the two measures coincide.

A foundational analysis shows that CE is *irreducible*: it is not derivable from any finitary or structural condition on the query system. The finite-cofinite counterexample and a model-theoretic argument via Łoś's theorem locate the boundary precisely. CE is not an assumption imposed from outside but the exact condition separating directed systems in which every convergence is witnessed somewhere in the refinement order from those in which it is not.

Beyond the extension theorem, the Stone space $\mathrm{St}(\mathcal{C})$ constructed here as a technical device has further significance: it is the canonical compact Hausdorff completion of the sample space for the observable topology. When an observation function is a cyclic vector for the Koopman operator of an underlying dynamical system, this Stone space is measure-theoretically isomorphic to the state space. The compact space introduced here as a technical device opens the possibility that the Stone space of the observable algebra is the state space itself — a connection made precise under reconstruction conditions developed elsewhere.

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1 Introduction

The programme of which this paper is a part does not begin with a probability space but with *distinctions*: the acts of observational separation by which an observer organises their experience of a system. State space, σ-algebra, and measure are not primitives — they are what coherent distinctions, pursued to their conclusion, are forced to produce. This paper proves the first step: a coherent family of observational distinctions determines a unique probability measure. The further steps — that this measure carries canonical dynamics, that the delay structure of observation (re)constructs the state space, and that finite data can certify (re)construction — are carried out in the companion papers.

The answer is given by a single condition. A compatible family of finitely additive charges on a directed system of Boolean algebras extends to a σ-additive measure at every level if and only if the family is *collectively exhaustive* (CE): for every decreasing sequence of events whose intersection is empty, some level in the refinement order witnesses the vanishing of mass. CE is not a structural condition on the query system but a condition on the charges — it rules out purely finitely additive charges, which assign persistent mass to sequences that vanish in the limit.

Two routes. The extension theorem is established by two independent proofs, each illuminating CE from a different direction.

The *Carathéodory route* (§4) assumes σ-additive marginals and constructs the global measure directly on the realization space. Sequential common refinements compress any countable cylinder family to a single level, reducing the σ-subadditivity check to a single marginal. By the CE theorem (§3), σ-additivity of the marginals is equivalent to CE.

The *Stone route* (§5) assumes only finite additivity and derives σ-additivity from compactness. Stone duality converts the directed Boolean system into an inverse system of compact Hausdorff spaces; a compatible family of charges assembles into a σ-additive measure on the inverse limit. Descending from the Stone space to Ω requires two conditions: Discriminability ensures the Stone embedding is injective, and CE appears as a support condition — the measure concentrates on the principal ultrafilters, i.e., on the image of Ω inside its Stone compactification.

CE is irreducible. CE is not derivable from any finitary or structural condition on the query system. The finite-cofinite content on $\mathbb{Q}$ satisfies every algebraic condition while failing CE; a model-theoretic argument via Łoś's theorem shows this is not a proof-search failure. CE is an irreducible commitment about the completed infinite, analogous to the axiom of infinity in arithmetic.

The Stone space. The Stone space $\mathrm{St}(\mathcal{C})$, constructed as a technical device for measure extension, is the canonical compact Hausdorff completion of Ω for the observable topology (§6). CE's two faces — algebraic convergence and support on Ω — are the same condition seen from inside and outside the algebra. When an observation function is cyclic for the Koopman operator, the Stone space is measure-theoretically isomorphic to the state space; this connection is developed in the companion papers.

Organisation. Section 2 introduces query systems, cylinder sets, the observable σ-algebra, compatible charges, and the three hypotheses (Surjective Evaluation, Discriminability, CE). Section 3 establishes the CE theorem and its irreducibility. Section 4 proves the Carathéodory route. Section 5 proves the Stone route. Section 6 draws the two routes together and discusses the Stone space.

2 Setup: query systems and charges

2.1 Observational structure

Definition 2.1 (Query system)**.** A *query system* is a tuple

$$(\iota, \leq, \{(\mathrm{Out}(i), \mathcal{F}_i)\}_{i\in\iota}, \{\pi_{ij}\}_{i\leq j}, \Omega, \{\mathrm{eval}_i\}_{i\in\iota})$$

consisting of:

- a nonempty partially ordered set $(\iota, \leq)$, the *index set*;
- for each $i \in \iota$, a measurable space $(\mathrm{Out}(i), \mathcal{F}_i)$, the *outcome space at level* i;
- for each $i \leq j$ in ι, a measurable surjection $\pi_{ij} : \mathrm{Out}(j) \to \mathrm{Out}(i)$, the *refinement map*;
- a set Ω, the *realization*, together with maps $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$ for each $i \in \iota$, the *evaluation maps*,

satisfying the *coherence condition*: $\pi_{ij} \circ \mathrm{eval}_j = \mathrm{eval}_i$ for all $i \leq j$.

The index set, outcome spaces, and refinement maps constitute the *observational structure* of the query system; they encode the distinctions the observer can make and how finer levels coarsen to coarser ones. The realization Ω and evaluation maps record how that structure is instantiated: Ω is the set of states consistent with the system, and coherence says evaluation is compatible with coarsening.

2.2 Realization and observables

The observational structure $(\iota, \leq, \{\mathrm{Out}(i)\}, \{\pi_{ij}\})$ determines a canonical realization: the projective limit

$$\Omega = \varprojlim_{i \in \iota} \mathrm{Out}(i) = \Big\{ x \in \prod_{i \in \iota} \mathrm{Out}(i) \;\Big|\; \pi_{ij}(x_j) = x_i \text{ whenever } i \leq j \Big\},$$

with $\mathrm{eval}_i(x) = x_i$.

Definition 2.2 (Common coarsenings and common refinements)**.** The index set $(\iota, \leq)$ admits *common coarsenings* if every pair $i, j \in \iota$ has a lower bound: there exists $k \leq i$ and $k \leq j$. It admits *common refinements* if every pair has an upper bound: there exists $k \geq i$ and $k \geq j$. It admits *sequential common refinements* if every countable family $\{i_n\}_{n \geq 1} \subset \iota$ has an upper bound in ι.

Sequential common refinements imply common refinements. Common coarsenings ensure that cylinder sets form a π-system (Lemma 2.4 below); sequential common refinements supply the σ-subadditivity step in the Carathéodory route.

Definition 2.3 (Cylinder sets)**.** For $i \in \iota$ and $A \in \mathcal{F}_i$, the *level-i cylinder with base A* is

$$\mathrm{Cyl}(i, A) \;=\; \{\omega \in \Omega : \mathrm{eval}_i(\omega) \in A\}.$$

The *cylinder generating family* is

$$\mathcal{C} \;=\; \{\mathrm{Cyl}(i, A) : i \in \iota,\ A \in \mathcal{F}_i\}.$$

For each $i \in \iota$, let B_i denote the Boolean algebra of subsets of Ω generated by $\{\mathrm{Cyl}(i, A) : A \in \mathcal{F}_i\}$. Then $\mathcal{C} = \bigcup_{i \in \iota} B_i$.

The *observable σ-algebra* is $\sigma(\mathcal{C})$, the σ-algebra on Ω generated by all cylinder sets.

The coherence condition gives, for $i \leq j$, $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$, so the same subset of Ω can be described as a cylinder at any finer level.

Lemma 2.4 (Cylinder π-system)**.** *Assume $(\iota, \leq)$ admits common coarsenings. Then $\mathcal{C}$ is a π-system: it is closed under finite intersections.*

Proof. Given $\mathrm{Cyl}(i, A)$ and $\mathrm{Cyl}(j, B)$, let k be a common lower bound, with $k \leq i$ and $k \leq j$. Then

$$\mathrm{Cyl}(i, A) \cap \mathrm{Cyl}(j, B) = \mathrm{Cyl}(k,\ \pi_{ki}^{-1}(A) \cap \pi_{kj}^{-1}(B)),$$

which is again a cylinder. ☐

For $i \leq j$, define the *connecting map*

$$\varphi_{ij} : B_i \to B_j, \qquad \varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A)).$$

This is a well-defined Boolean algebra homomorphism. The family $\{B_i, \varphi_{ij}\}$ is a directed system of Boolean algebras.

2.3 Charges and hypotheses

Definition 2.5 (Compatible family)**.** A *charge* on a Boolean algebra B is a finitely additive function $\mu : B \to [0, \infty)$ with $\mu(\emptyset) = 0$. A family of charges $\{\mu_i\}_{i\in\iota}$ with μ_i on B_i is *compatible* if

$$\mu_j(\varphi_{ij}(E)) = \mu_i(E) \qquad \text{for all } i \le j \text{ and all } E \in B_i.$$

A compatible family also determines a charge on $\mathcal{C}$ by $\mu(\mathrm{Cyl}(i, A)) := \mu_i(\mathrm{Cyl}(i, A))$; compatibility ensures this is well-defined.

The main theorems require three conditions.

Definition 2.6 (Surjective Evaluation)**.** The query system satisfies *Surjective Evaluation* if $\mathrm{eval}_i : \Omega \to \mathrm{Out}(i)$ is surjective for every $i \in \iota$.

Definition 2.7 (Discriminability)**.** The query system *discriminates points* if for every $\omega \neq \omega'$ in Ω there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$.

Definition 2.8 (Collective Exhaustion)**.** A charge μ on $\mathcal{C}$ (the common extension of a compatible family $\{\mu_i\}$) satisfies *collective exhaustion* (CE) if: whenever $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{C}$ and $\bigcap_n E_n = \emptyset$, we have $\mu(E_n) \to 0$.

Surjective Evaluation is a structural condition ensuring that every outcome at every level is realized by some element of Ω; it implies that the refinement maps π_{ij} are surjective (by coherence) and that the connecting maps φ_{ij} are injective. Discriminability ensures the Stone embedding $\omega \mapsto \{\text{principal ultrafilter at } \omega\}$ is injective. CE is the irreducible valuation-layer condition studied in §3; it rules out purely finitely additive charges and is the necessary and sufficient condition for σ-additive extensibility.

3 The CE theorem

The argument of §§3–5 has a single subject: the single condition under which a coherent family of observations determines a probability measure. It appears in four forms — as a local continuity criterion (§3, single algebra), as a global equivalence across the directed system (§3.3), as a direct measure-theoretic construction (§4), and as a topological derivation via compactness (§5). These are not separate results but four projections of a single constraint; §6 identifies the point where all four coincide.

This section establishes collective exhaustion as the necessary and sufficient condition for σ-additive extensibility. The analysis works at the level of an abstract directed system of Boolean algebras; it does not require the full query-system structure of §2. To keep this generality clean, we write $\mathcal{E}_i$ for a generic Boolean algebra at level i and ℓ_i for the charge on it. The connection to query systems is: $\mathcal{E}_i = B_i$ (the Boolean algebra of cylinder sets at level i), $O_i = \mathrm{Out}(i)$, and $\ell_i = \mu_i$. The full query system structure re-enters in §4 and §5.

3.1 Single-algebra extension

Definition 3.1 (Boolean charge space)**.** A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where $\mathcal{E}$ is a Boolean algebra of subsets of a set O and $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation. No σ-algebra and no topology are assumed on O.

Under Stone's representation theorem [11], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected Hausdorff space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The *gap* $\sigma(\mathcal{E}) \setminus \mathcal{E}$ consists of the Baire sets that are not clopen: events that countable operations can reach but that no single observable distinction can name.

Theorem 3.2 (Extension criterion). *A normalized finitely-additive $\ell : \mathcal{E} \to [0, 1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_n E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$. The extension is then unique.*

Proof. Standard; see Halmos [6] or Fremlin [5] Vol. 1. Continuity at $\emptyset$ is used in the Carathéodory extension argument to pass from finite additivity to σ-additivity on $\sigma(\mathcal{E})$. $\square$

Remark 3.3 (The gap and the regress). The condition in Theorem 3.2 is not verifiable from within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. In a directed system, one might hope a finer level $j \geq i$ can witness the emptiness that level i cannot see — but the obstruction reappears: finite additivity at j does not force continuity at $\emptyset$ for sequences in $\sigma(\mathcal{E}_j)$. Something external to any single level is required, and §3.2 shows the directed-system structure alone does not supply it.

3.2 Nontrivial refinement

Definition 3.4 (Nontrivial refinement). Let $\pi : O_j \to O_i$ be a surjective refinement map, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$.

Proposition 3.5 (Nontrivial refinement introduces new events). *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. Equivalently, a level i at which $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$ is one above which the refinement order is algebraically trivial: no finer level adds new discriminative power.*

Proof. Nontriviality asserts the existence of $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$, directly giving the strict inclusion. The contrapositive is the stated equivalence. $\square$

Proposition 3.6 (Gap nonemptiness). *Let $\mathcal{E}$ be a Boolean algebra of subsets of a countably infinite set O that is not a σ-algebra. Then $\sigma(\mathcal{E}) \setminus \mathcal{E} \neq \emptyset$.*

Proof. If $\mathcal{E}$ is not a σ-algebra, there exists a countable family $\{A_n\} \subset \mathcal{E}$ with $\bigcup_n A_n \notin \mathcal{E}$. By definition $\bigcup_n A_n \in \sigma(\mathcal{E})$, so it lies in the gap. $\square$

3.3 CE characterises σ-additivity

We now state the directed-system version of CE and prove the main equivalence.

Definition 3.7 (Collectively exhaustive family). A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$.

Here $\pi_{ij} : O_j \to O_i$ is the refinement map (j finer than i), so $\pi_{ij}^{-1}(E_n) \subseteq O_j$ is the preimage of $E_n \subseteq O_i$ at the finer level.
CE is a *valuation-layer* condition: it places a requirement on how the charges behave, not on the Boolean-algebraic index structure. It says that no level can assign persistent mass to events that the system as a whole sees as empty.

The following proposition establishes that the index-layer structure of a directed system — sequential upper-directedness and charge compatibility — does not on its own force σ-additivity.

Proposition 3.8 (Index-layer conditions do not force σ-additivity). *There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. Let $\iota = \mathbb{N} \cup \{\omega\}$ with $n \leq \omega$ for all $n \in \mathbb{N}$, directed by this order. Take all outcome spaces to be $\mathbb{Q}$, all event algebras to be the finite-cofinite algebra $\mathcal{E} = \{E \subseteq \mathbb{Q} : E \text{ is finite or } E^c \text{ is finite}\}$, all refinement maps to be the identity, and $\ell(E) = 0$ if E is finite, $\ell(E) = 1$ if E is cofinite. This is a normalized finitely-additive content and compatibility holds trivially.

Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$ and let $F_n = \{q(0), \dots, q(n)\}$. Each F_n is finite so $\ell(F_n) = 0$, yet $\bigcup_n F_n = \mathbb{Q}$ and $\ell(\mathbb{Q}) = 1$. σ-subadditivity would give $1 \leq \sum_n \ell(F_n) = 0$: contradiction. $\square$

Theorem 3.9 (CE Characterisation Theorem). *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By CE, find $j \geq i$ with $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. Theorem 3.2 gives the unique σ-additive extension.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every i. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization. Take $j = i$. $\square$

Remark 3.10 (Dissolution of the index-valuation conflation). The question "do the structural conditions on a directed system force σ-additivity?" is ill-posed: it conflates conditions on the index structure with conditions on the charges. Proposition 3.8 separates them; Theorem 3.9 identifies the exact valuation-layer condition. Sequential upper-directedness enters only when assembling per-level extensions into a global measure (§4), not at the per-level step.

3.4 CE is irreducible

Proposition 3.11 (CE irreducibility). *Collective exhaustion is not implied by any structural condition on a query system.*

(i) (Particular case.) *Sequential upper-directedness, compatibility, and normalization do not imply CE.*

(ii) (Universal case.) *No condition Φ expressible in the first-order language of query systems implies CE.*

Proof. The particular case is Proposition 3.8: the finite-cofinite content satisfies all named structural hypotheses and fails CE.

For the universal case, we proceed in three steps.

Step 1. A first-order condition on query systems is preserved under ultraproducts by Łoś's theorem [9]: if Φ holds in every member of a family of query systems, it holds in their ultraproduct.

Step 2. Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$ extending the cofinite filter, and let δ_n be the point mass at $q(n) \in \mathbb{Q}$ (using the enumeration from Proposition 3.8). Each δ_n is a normalized σ-additive content satisfying every structural condition including CE. The ultraproduct $\prod_{\mathcal{U}} \delta_n$ assigns to $E \subseteq \mathbb{Q}$ the value $\lim_{\mathcal{U}} \mathbf{1}_{q(n) \in E}$, which is 1 on cofinite sets and 0 on finite sets — exactly the finite-cofinite content ℓ from Proposition 3.8.

Step 3. By Step 1, any universally valid first-order Φ holds for ℓ. But ℓ fails CE. Therefore no first-order Φ can imply CE.

The Yosida–Hewitt decomposition [13] explains why this obstruction is permanent. Every finitely-additive probability on a Boolean algebra decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive*: it satisfies every algebraic condition while assigning zero to every singleton. The finite-cofinite content ℓ is purely finitely additive ($\ell_c = 0$). CE is exactly the condition that rules out a nonzero purely finitely additive part: $\ell_p = 0$. $\square$

Remark 3.12 (CE as irreducible primitive)**.** The gap between structural coherence and σ-additivity is a proved boundary, not a proof-search failure. CE is analogous to the axiom of infinity: not derivable from the remaining structure, not refutable, but the commitment without which probability cannot proceed. The ultrafilter-based content is the canonical witness: it passes every finite consistency test yet assigns unit mass to events whose infinite intersection is empty.

CE is the *principle of witnessed vanishing*: a distinction may disappear globally only if that disappearance is witnessed at some finite level of the refinement order. No global collapse without local evidence.

4 Route 1: The Carathéodory extension

This section establishes the extension theorem by a direct Carathéodory argument. This is the third face of the argument: the measure constructed directly from within the algebra. The input is a query system satisfying Surjective Evaluation, sequential common refinements, and common coarsenings, together with a compatible family of σ-additive marginals. By Theorem 3.9, the σ-additivity of the marginals is equivalent to CE applied at each level.

4.1 Observational determination

Proposition 4.1 (Observational determination)**.** *Let the query system admit common coarsenings and let P, P' be probability measures on $(\Omega, \sigma(\mathcal{C}))$. If $(\mathrm{eval}_i)_\# P = (\mathrm{eval}_i)_\# P'$ for all $i \in \iota$, then $P = P'$.*

Proof. The equality of pushforward laws implies P and P' agree on every cylinder event. By Lemma 2.4 the cylinder family $\mathcal{C}$ is a π-system generating $\sigma(\mathcal{C})$. The π–λ theorem gives $P = P'$. □

4.2 Observable content

Lemma 4.2 (Finite observational content)**.** *Assume the query system admits common coarsenings and common refinements and satisfies Surjective Evaluation, and let $\{\nu_i\}_{i\in\iota}$ be an observationally compatible family with each $\nu_i \in \mathcal{P}(\mathrm{Out}(i))$. Then the assignment*

$$\mu_0(\mathrm{Cyl}(i, A)) := \nu_i(A), \qquad A \in \mathcal{F}_i,$$

defines a well-defined finitely additive content on $\mathcal{C}$.

Proof. Well-definedness. Suppose $\mathrm{Cyl}(i, A) = \mathrm{Cyl}(j, B)$. Use common refinements to find k with $k \geq i$ and $k \geq j$. The constraint sets in $\mathrm{Out}(k)$ are $\pi_{ik}^{-1}(A)$ and $\pi_{jk}^{-1}(B)$. The cylinder equality and Surjective Evaluation (surjectivity of eval_k) imply these coincide in $\mathrm{Out}(k)$. Compatibility then gives $\nu_i(A) = \nu_k(\pi_{ik}^{-1}(A)) = \nu_k(\pi_{jk}^{-1}(B)) = \nu_j(B)$.

Finite additivity. For disjoint cylinders at possibly different levels, compress to a common refinement k and use additivity of ν_k. □

4.3 Carathéodory extension theorem

Theorem 4.3 (Observational extension theorem)**.** *Suppose:*

(i) $(\iota, \leq)$ admits common coarsenings and sequential common refinements;

(ii) the query system satisfies Surjective Evaluation;

(iii) the family $\{\nu_i\}_{i\in\iota}$ is observationally compatible and each ν_i is a probability measure on $(\mathrm{Out}(i), \mathcal{F}_i)$.

Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that $(\mathrm{eval}_i)_\# P = \nu_i$ for every $i \in \iota$.

Proof. Step 1: Finite content. Theorem 4.2 gives a well-defined finitely additive content μ_0 on $\mathcal{C}$.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(i, B) \subseteq \bigcup_{n \geq 1} \mathrm{Cyl}(i_n, A_n)$. Apply sequential common refinements to $\{i, i_1, i_2, \ldots\}$ to find a single index k with $k \geq i$ and $k \geq i_n$ for all n. Compress each cylinder to level k:

$$\mathrm{Cyl}(i, B) = \mathrm{Cyl}(k, E), \qquad \mathrm{Cyl}(i_n, A_n) = \mathrm{Cyl}(k, E_n),$$

where $E = \pi_{ik}^{-1}(B)$ and $E_n = \pi_{i_n k}^{-1}(A_n)$. Surjective Evaluation and the cylinder inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in $\mathrm{Out}(k)$. Therefore

$$\mu_0(\mathrm{Cyl}(i, B)) = \nu_k(E) \leq \sum_{n=1}^{\infty} \nu_k(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(i_n, A_n)),$$

using σ-subadditivity of the single measure ν_k.

Step 3: Carathéodory extension. The family $\mathcal{C}$ is a semiring generating $\sigma(\mathcal{C})$ and μ_0 is a σ-subadditive finitely additive content on it. Carathéodory's extension theorem [1, 2] applies and yields P on $(\Omega, \sigma(\mathcal{C}))$.

Step 4: Marginal recovery and normalization. For any $A \in \mathcal{F}_i$, $P(\mathrm{Cyl}(i, A)) = \mu_0(\mathrm{Cyl}(i, A)) = \nu_i(A)$, so $(\mathrm{eval}_i)_\# P = \nu_i$. Setting $A = \mathrm{Out}(i)$ gives $P(\Omega) = 1$.

Uniqueness. Two probability measures with the same evaluation marginals agree on $\mathcal{C}$, hence on $\sigma(\mathcal{C})$ by Theorem 4.1. □

Remark 4.4 (Relation to Kolmogorov's extension theorem)**.** Kolmogorov's theorem [8] constructs a probability measure on S^T from consistent finite-dimensional distributions. Theorem 4.3 plays the same role for query systems: the realization space $\Omega = \varprojlim \mathrm{Out}(i)$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition. The key conceptual difference is that no topology is imposed on the outcome spaces: the measure P is built directly on the measurable structure of Ω by Carathéodory, without any appeal to tightness or compactness.

Remark 4.5 (Prokhorov route)**.** A third route to the extension theorem starts from *finitely* additive marginals and derives σ-additivity from topological compactness. Under *surjective projective uniform tightness* — a condition requiring that for every $\varepsilon > 0$ there exist compact sets $K_i \subset \mathrm{Out}(i)$ with $\nu_i(\mathrm{Out}(i) \setminus K_i) < \varepsilon$ and $\pi_{ij}(K_j) = K_i$ for all $i \leq j$ — the Prokhorov theorem extends finitely additive marginals on a countable sequentially upper-directed query system with Hausdorff outcome spaces to a σ-additive global measure. This route assumes topological structure on the outcome spaces and is complementary to the Carathéodory and Stone routes, which require no topology.

Remark 4.6 (Lean formalization)**.** Theorem 4.3 is fully formalized in Lean 4 (`observational_extension` in `QuerySystem.lean`, 0 sorrys). The hypotheses correspond exactly to the four conditions: `LowerDirected`, `SequentiallyUpperDirected`, `EvalSurjective`, and `CompatibleMarginals` with `IsProbabilityMeasure`. The Carathéodory step uses `AddContent.measure` from Mathlib [12].

5 Route 2: The Stone extension

This section gives the second, independent proof of the extension theorem. This is the fourth face: the same measure derived from outside, via compactness. The input is weaker: charges need only be finitely additive. σ-additivity is derived from the compactness of the Stone space.

The proof proceeds in four steps. Step A identifies the Stone space of the direct limit with the inverse limit of individual Stone spaces. The inverse limit measure step uses Choksi's theorem to produce a σ-additive measure on the inverse limit. Step B1 identifies the pullback of the Borel algebra of the Stone space with the observable σ-algebra. Step B2 shows that CE forces the measure to concentrate on the image of Ω.

5.1 Step A: Stone space of the direct limit

Lemma 5.1 (Injectivity of connecting maps)**.** *If the query system satisfies Surjective Evaluation, then $\varphi_{ij} : B_i \to B_j$ is an injective Boolean algebra homomorphism for every $i \leq j$.*

Proof. Surjective Evaluation and the coherence condition give $\pi_{ij}(\mathrm{eval}_j(\omega)) = \mathrm{eval}_i(\omega)$ for all ω, so $\mathrm{im}(\pi_{ij}) \supseteq \mathrm{im}(\mathrm{eval}_i) = \mathrm{Out}(i)$: the refinement maps are surjective. A surjective map π_{ij} has injective preimage map π_{ij}^{-1} on subsets, so $\varphi_{ij}(\mathrm{Cyl}(i, A)) = \mathrm{Cyl}(j, \pi_{ij}^{-1}(A))$ is injective on generators, hence on B_i. □

Proposition 5.2 (Step A)**.** *The Stone space of the direct limit satisfies* $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$.

Proof. By Lemma 5.1, the connecting maps φ_{ij} are injective Boolean algebra homomorphisms. Under Stone duality, injective Boolean algebra homomorphisms correspond to surjective continuous maps between Stone spaces. The Stone space functor St converts colimits in the category of Boolean algebras to limits in the category of compact Hausdorff spaces [7, II.4]. Applied to $\bigcup_i B_i$ with its injective system, this gives $\mathrm{St}\big(\bigcup_i B_i\big) \cong \varprojlim_i \mathrm{St}(B_i)$. □

5.2 The inverse limit measure

Proposition 5.3 (Inverse limit measure)**.** *Let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$. Then there exists a unique probability measure $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$ whose pushforward along each projection $\varprojlim_i \mathrm{St}(B_i) \to \mathrm{St}(B_j)$ agrees with the Borel measure on $\mathrm{St}(B_j)$ corresponding to μ_j.*

Proof. Step 1: Single-algebra correspondence. For each i, Stone duality identifies B_i with the clopen algebra $\mathrm{Clop}(\mathrm{St}(B_i))$. A charge μ_i on B_i thus defines a finitely additive set function on the clopens of the compact Hausdorff space $\mathrm{St}(B_i)$. Since $\mathrm{St}(B_i)$ is compact and totally disconnected, the clopens form a base for the topology and a finitely additive charge on them extends uniquely to a regular Borel measure; see [3], §6, or [6], §§53–54. Call this measure $\hat{\mu}_i$ on $\mathrm{St}(B_i)$.

Step 2: Compatibility. The compatibility condition $\mu_j(\varphi_{ij}(E)) = \mu_i(E)$ translates, under Stone correspondence, to $\hat{\mu}_i$ being the pushforward of $\hat{\mu}_j$ along the bonding map $\mathrm{St}(B_j) \to \mathrm{St}(B_i)$. The family $\{\hat{\mu}_i\}$ is compatible on the inverse system $\{\mathrm{St}(B_i)\}$.

Step 3: Extension to the inverse limit. Each $\mathrm{St}(B_i)$ is compact Hausdorff. The inverse limit $\varprojlim_i \mathrm{St}(B_i)$ is compact Hausdorff. By [4, Theorem 1], a compatible family of regular Borel probability measures on a cofiltered inverse system of compact Hausdorff spaces with surjective bonding maps extends uniquely to a regular Borel probability measure on the inverse limit. This gives $\hat{\mu}$ on $\mathcal{B}(\varprojlim_i \mathrm{St}(B_i))$. □

5.3 Step B1: the structural identification

Let $\mathrm{pure} : \Omega \to \mathrm{St}(\mathcal{C})$ denote the Stone embedding, which sends $\omega \in \Omega$ to its principal ultrafilter $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$. For $E \in \mathcal{C}$, write $[E] = \{u \in \mathrm{St}(\mathcal{C}) : E \in u\}$ for the corresponding basic clopen in $\mathrm{St}(\mathcal{C})$.

Proposition 5.4 (Step B1: structural identification)**.** *Suppose the query system discriminates points. Then*

$$\mathrm{pure}^{-1}\big(\mathcal{B}_{\mathrm{Ba}}(\mathrm{St}(\mathcal{C}))\big) \;=\; \sigma(\mathcal{C}),$$

where $\mathcal{B}_{\mathrm{Ba}}$ denotes the Baire σ-algebra.

Proof. **($\supseteq$):** For any $E = \mathrm{Cyl}(i, A) \in \mathcal{C}$, $\mathrm{pure}^{-1}([E]) = \{\omega : E \in \mathrm{pure}(\omega)\} = \{\omega : \omega \in E\} = E$. Since $[E]$ is clopen, hence Baire, every cylinder set is in $\mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$, and therefore $\sigma(\mathcal{C}) \subseteq \mathrm{pure}^{-1}(\mathcal{B}_{\mathrm{Ba}})$.

($\subseteq$): The Stone space $\mathrm{St}(\mathcal{C})$ is compact and totally disconnected, so its clopens form a topological basis. The clopens are exactly the sets $[E]$ for $E \in \mathcal{C}$ [7, I.3.2], so the Baire σ-algebra of $\mathrm{St}(\mathcal{C})$ is $\sigma(\{[E] : E \in \mathcal{C}\})$. For any Baire set V, $\mathrm{pure}^{-1}(V) \in \sigma(\{\mathrm{pure}^{-1}([E]) : E \in \mathcal{C}\}) = \sigma(\mathcal{C})$.

Injectivity. Discriminability implies pure is injective: if $\omega \neq \omega'$, there exists $E \in \mathcal{C}$ with $\omega \in E$ and $\omega' \notin E$, so $E \in \mathrm{pure}(\omega)$ but $E \notin \mathrm{pure}(\omega')$. Injectivity ensures the pullback does not conflate distinct events. □

Remark 5.5. On a compact Hausdorff space the Baire σ-algebra is contained in the Borel σ-algebra, with equality when the space is second-countable. We work throughout with the Baire σ-algebra on $\mathrm{St}(\mathcal{C})$; this suffices because the measure $\hat{\mu}$ constructed above is a Baire measure (it is the unique extension of a charge on the clopen algebra), and Choksi's theorem applies to Baire measures on compact spaces.

5.4 Step B2: CE and the support condition

Proposition 5.6 (Step B2: support condition)**.** *Let μ be a charge on $\mathcal{C}$ satisfying CE. Then the corresponding Baire measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is supported on $\mathrm{pure}(\Omega)$, the set of principal ultrafilters.*

Proof. By the Yosida–Hewitt theorem [13, 10], every bounded charge on a Boolean algebra decomposes uniquely as $\mu = \mu_c + \mu_p$, where μ_c is countably additive and μ_p is purely finitely additive. Under Stone duality, μ_c corresponds to mass on the principal ultrafilters $\mathrm{pure}(\Omega) \subset \mathrm{St}(\mathcal{C})$, while μ_p corresponds to mass on the non-principal ultrafilters [10], Ch. 10.

CE is equivalent to $\mu_p = 0$: CE requires $\mu(E_n) \to 0$ whenever $E_n \searrow \emptyset$, which is exactly the condition that the purely finitely additive part vanishes (Theorem 3.9). Therefore $\hat{\mu}_p = 0$ and $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. □

5.5 The Stone extension theorem

Theorem 5.7 (Stone extension theorem)**.** *Let the query system satisfy Surjective Evaluation, Discriminability, and common coarsenings, and let $\{\mu_i\}$ be a compatible family of normalised charges on $\{B_i\}$ satisfying CE. Then there exists a unique probability measure P on $(\Omega, \sigma(\mathcal{C}))$ such that*

$$P(\mathrm{Cyl}(i, A)) \;=\; \mu_i(\mathrm{Cyl}(i, A)) \qquad \textit{for all } i \in \iota \textit{ and } A \in \mathcal{F}_i.$$

Proof. The proof assembles the preceding propositions:

$$\{\mu_i\} \xrightarrow{\S 5.2,\,\mathrm{Step}\ 1} \{\hat{\mu}_i\} \xrightarrow{\S 5.2,\,\mathrm{Step}\ 3} \hat{\mu} \text{ on } \varprojlim_i \mathrm{St}(B_i) \xrightarrow{\mathrm{B1}} P \text{ on } \sigma(\mathcal{C}).$$

Existence. By Proposition 5.2, $\mathrm{St}(\mathcal{C}) \cong \varprojlim_i \mathrm{St}(B_i)$. By Proposition 5.3, the compatible family $\{\mu_i\}$ extends to a Baire probability measure $\hat{\mu}$ on $\varprojlim_i \mathrm{St}(B_i) \cong \mathrm{St}(\mathcal{C})$.

By Proposition 5.6, $\hat{\mu}$ is supported on $\mathrm{pure}(\Omega)$. Since pure is injective (Discriminability), the pushforward $P = (\mathrm{pure})_*\hat{\mu}$ — the image measure under pure^{-1} on the support — is a well-defined probability measure on Ω.

By Proposition 5.4, pure^{-1} maps Baire sets in $\mathrm{St}(\mathcal{C})$ to sets in $\sigma(\mathcal{C})$, so P is defined on $\sigma(\mathcal{C})$. For any cylinder set $E = \mathrm{Cyl}(i, A)$,

$$P(E) \;=\; \hat{\mu}([E]) \;=\; \mu_i(\mathrm{Cyl}(i, A)).$$

Uniqueness. Two probability measures on $(\Omega, \sigma(\mathcal{C}))$ agreeing on all cylinder sets are equal by Theorem 4.1, which requires common coarsenings. The Stone theorem therefore assumes common coarsenings in addition to Surjective Evaluation, Discriminability, and CE. $\square$

Remark 5.8 (CE is not bypassed by compactness)**.** It may appear that the compactness of the Stone space eliminates the need for CE. Compactness does ensure that $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$ is σ-additive. But $\hat{\mu}$ lives on the Stone space, not on Ω. The descent from $\mathrm{St}(\mathcal{C})$ to Ω requires that no mass escapes to the non-principal ultrafilters, and that is precisely CE (Proposition 5.6). In the Stone picture, CE is not an additional hypothesis but a *support condition*: the measure concentrates on the image of the sample space inside its Stone compactification.

Remark 5.9 (Lean formalization)**.** The Stone route is formalized in Lean 4 in `StoneDualityExtension.lean`. The structural results (Tasks 0′-A and 0′-C: Stone space, cylinder clopens, evaluation maps, bonding maps) are fully proved with no sorry obligations. Two intentional sorrys mark Mathlib gaps: the extension of a clopen-algebra charge to a regular Borel measure on a compact totally-disconnected space (Halmos §§53–54; Fremlin Vol. 1 §311E), and Choksi's projective-limit theorem for compatible measures on cofiltered inverse systems of compact Hausdorff spaces. The coincidence of routes (Task 0′-E: `stone_agrees_with_caratheodory`) is fully proved, as an immediate consequence of `observational_determination`.

6 The bridge

6.1 Two routes, one theorem

Under shared hypotheses — Surjective Evaluation, Discriminability, and σ-additive compatible marginals — both routes produce the same measure.

Proposition 6.1 (Coincidence of routes)**.** *Suppose the query system satisfies Surjective Evaluation, Discriminability, and sequential common refinements and common coarsenings, and let $\{\nu_i\}$ be a compatible family of probability measures. Let P^{C} be the measure produced by Theorem 4.3 and P^{S} the measure produced by Theorem 5.7. Then $P^{\mathrm{C}} = P^{\mathrm{S}}$.*

Proof. Both measures agree on every cylinder set: $P^{\mathrm{C}}(\mathrm{Cyl}(i, A)) = \nu_i(A) = P^{\mathrm{S}}(\mathrm{Cyl}(i, A))$. By observational determination (Theorem 4.1), they are equal on $\sigma(\mathcal{C})$. $\square$

The two routes are genuinely independent proofs of the same theorem. The Carathéodory route works directly within measure theory; the Stone route takes a detour through topology, using compactness of the Stone space to derive σ-additivity that the Carathéodory route assumes. That they produce the same measure is not obvious from the constructions alone; it is a consequence of uniqueness (Theorem 4.1), which is doing real work here.

6.2 CE's two faces

CE appears in the two routes in structurally different forms, yet it is the same condition.

In the Carathéodory route (Route 1), CE appears at the level of the charges: the per-level marginals are σ-additive, which by Theorem 3.9 is equivalent to CE applied at each level. Here, CE is an *algebraic* condition: the charges do not assign persistent mass to events that the observer's refinement hierarchy sees as empty. It is a condition on the observer's commitments.

In the Stone route (Route 2), CE appears as a *topological* condition: the Baire measure $\hat{\mu}$ on the Stone space must be supported on $\mathrm{pure}(\Omega)$, the image of the sample space inside its compact compactification. Mass on non-principal ultrafilters corresponds exactly to the purely finitely additive part μ_p, and CE $\Leftrightarrow$ $\mu_p = 0$ (Proposition 5.6). Here, CE is a *geometric* condition: the observer's measure does not escape to the "phantom" points of the Stone compactification.

The Yosida–Hewitt decomposition is the bridge between the two faces: it identifies the purely finitely additive part of μ with the mass that has escaped to non-principal ultrafilters. CE says both things simultaneously: the algebraic convergence is honest, and the geometric measure is confined to Ω.

6.3 The Stone space as compact completion

The Stone space $\mathrm{St}(\mathcal{C})$ constructed in §5 as a technical device for measure extension is the canonical compact Hausdorff completion of the sample space for the observable topology. The Stone embedding pure : $\Omega \to \mathrm{St}(\mathcal{C})$ is dense — every nonempty basic clopen $[E]$ meets pure(Ω) — and the Stone space is uniquely determined by $\mathcal{C}$ up to homeomorphism. It is not imposed on Ω from outside but the compactification already contained in the Boolean algebra of cylinder sets.

Every point in $\mathrm{St}(\mathcal{C})$ is an ultrafilter: a maximal consistent assignment of yes/no answers to every observable distinction. Principal ultrafilters correspond to actual sample-space points; non-principal ultrafilters are the ideal limit points — consistent distinction patterns the refinement system is coherent with, but that no element of Ω realises. The Stone space is therefore the *space of all consistent distinction patterns*: the completion of the observer's world to a compact whole.

CE is the condition that closes the loop. It requires the measure P on Ω, when extended to $\mathrm{St}(\mathcal{C})$ via pure, to remain concentrated on the realised part: no mass escapes to the phantom ultrafilters. An observer satisfying CE retains no phantom mass: probability remains confined to the realised world, however far the algebraic completion extends. The Stone topology, generated by all cylinder clopens, is the coarsest topology making every observable distinction continuous; in this sense the Boolean framework is a first approximation, and effective separation is already graded by the measure — a refinement developed in the companion papers.

For a dynamical system (X, T, μ) with observation function h, the time-delay cylinder algebras form a directed system of exactly the kind treated here. The Stone space of the resulting observable algebra is, when h is cyclic for the Koopman operator U_T, measure-theoretically isomorphic to the state space X itself. The compact space introduced here as a tool opens the possibility that the Stone space of the observable algebra *is* the state space — a connection made precise under (re)construction conditions developed in the companion papers.

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